

Class 9 AI Project Assignment

Project Title: ATM Machine Simulator using Python Functions

Objective:

Create a Python program to simulate a simple ATM system that includes PIN verification, and basic banking operations.

Problem Statement:

You are required to design an ATM Machine Simulator program with the following functionalities:

1. PIN Verification:

- The program should ask the user to enter a 4-digit PIN.
- The user gets 3 attempts to enter the correct PIN.
- If the PIN is correct, access is granted.
- If all attempts fail, the program should display "Card blocked" and exit.

2. ATM Operations Menu:

After successful PIN verification, display a menu with these options:

- Check Balance – Display current account balance.
- Deposit Money – Allow the user to deposit money and update balance.
- Withdraw Money – Allow the user to withdraw money if sufficient balance is available.
- Exit – Exit the program.

3. The starting account balance can be ₹5000.

4. The program should loop through the menu until the user chooses to exit.

5. Each main task must be implemented as a separate function.

Suggested Functions to Implement:

- verify_pin(): Ask the user for the PIN, allow 3 attempts, and return True/False for access.
- check_balance(balance): Display the current balance.
- deposit(balance): Ask for an amount, add it to balance, and return updated balance.
- withdraw(balance): Ask for an amount, subtract it from balance if enough money, return updated balance.
- main(): Main function to control program flow: PIN verification, menu display, and calling other functions.

Instructions for Students:

1. Write your Python code for the ATM simulator using functions.
2. Run the program and take screenshots or print the outputs of each operation (Check Balance, Deposit Money, Withdraw Money, Exit).
3. Submit a hard copy containing:
 - Printed Python code
 - Printed program output showing all functionalities
 - Any comments or explanation you added in code
4. Deadline: 5 November
5. Marks: 10

Optional Features (for Extra Credit):

- Display last transaction details.
- Allow the user to change the PIN after successful login.
- Add transaction IDs for each deposit or withdrawal.